ENISOLAR Rover — One-Week Demo Plan

A simple three-person university project

From available parts to a safe, local rover demonstration

Planning baseline · 5 November 2026

Goal. Low-speed rover control from a simple local dashboard, a visible status display, and one repeatable navigation demonstration if hardware allows. **Limit.** This is a class prototype, not a secure production rover or a certification.

Contents

1 One-week scope 1
2 Person-by-person checklist 1
2.1 Shared daily rule 2
3 Six working docs 2
4 Safety and acceptance 2
5 Glossary 2

1 One-week scope

Start from the parts the team can obtain immediately. Use standard development hardware and an existing chassis if available; avoid custom PCB, new cloud services, machine-learning detection, OTA, and advanced autonomy this week. If the physical build is blocked, use a simulator or mocked status and say so clearly.

Use one local connection only: a local Wi-Fi access point/web page is the simple default if supported; use BLE only if that path already works. Keep speed low, test stop/disconnect behavior, and do not expose control to the public internet. A short timed route or small waypoint sequence is an optional demo, not robust autonomous navigation.

Day	Milestone	Work	Exit check
1	Scope and parts	Confirm parts, safe pin/power map, local connection, command names and demo area.	Parts and risks known; no unreviewed wiring.
2	Safe bring-up	Logic power and controller first; test motor driver with wheels lifted; check physical disable.	Boot/reset keeps motors off; stop path tested.
3	Core features	Bounded drive commands, dashboard status, and simple manual or timed navigation.	Commands and stop work on real hardware or simulator.
4	Integration	Test invalid input, disconnect, reconnect, reboot, stop and demo flow together.	No critical demo bug; fallback ready.
5	Freeze and present	Feature freeze, three run-throughs, setup guide, limitations and demo script.	Repeatable, safe demonstration.

2 Person-by-person checklist

Person	Owns these docs	Daily to-do
1	Wiring schematic; Firmware	Day 1 inventory and pin/power map. Day 2 safe wiring, motor driver and disable test. Day 3 boot, bounded motor commands, status and stop-on-timeout. Day 4 integrate. Day 5 final build and recovery steps.
2	Navigation; Connectivity	Day 1 choose one local link and demo behavior. Day 2 agree command/status format. Day 3 manual control and a simple timed path or simulated waypoint. Day 4 disconnect/reconnect and invalid-input tests. Day 5 document connection and limits.
3	Dashboard; Security	Day 1 scope board and UI sketch. Day 2 stop button and connection state. Day 3 allowed controls and basic checks. Day 4 lead joint integration test. Day 5 rehearse demo and review docs/limitations.

2.1 Shared daily rule

Agree one deliverable per person each morning and integrate by the end of the day. Keep a known-good version. If parts, drivers, or connection fail on Day 1, switch early to the simulator or simpler local control rather than waiting until the final day.

3 Six working docs

Document	Purpose for this demo
Wiring schematic	Actual pin/power map, component ratings, motor disable and bring-up checklist. Fill from the parts on hand.
Firmware	Small boot, command allowlist, speed bounds, motor control, status and timeout stop implementation.
Navigation	Manual control first; one timed route or simple waypoint demonstration only if sensors are ready.
Connectivity	One local link, tiny command/status format and disconnect behavior. No public internet.
Dashboard	Connection/mode/fault display, basic controls, and prominent stop button.
Security	No committed secrets, local-only link, input checks, stop/disconnect test and honest limitations.

4 Safety and acceptance

Test with wheels lifted first, low speed, a clear bounded area, a spotter, and an accessible physical power cut-off or emergency stop (E-stop). Never operate near roads, stairs, people, or fragile equipment. Stop immediately on unexpected motion or a failed stop. Firmware stop is not a substitute for a physical stop. The demo is ready when the rover (or clearly labelled simulator) starts reliably, accepts only the small documented command set, displays connection/status, stops on command and on disconnect, and completes the same simple demo route three times. Document unimplemented features. The supplied slides' detector metrics, security claims, and example timeout values are not validated requirements for this class project.

5 Glossary

***BLE* – Bluetooth Low Energy**: An optional local phone-to-rover link.

***E-stop* – emergency stop**: A physical stop or motor-disable control, where supported by the available hardware.

***Firmware* – embedded software**: Software running on the rover controller.

***MVP* – minimum viable prototype**: The smallest working demonstration agreed by the team.

***Wi-Fi* – local wireless network**: The suggested local link for a simple dashboard demo.